Generative AI for Data Science Applications – 960290

Assignment 2

Group 1 :

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Following our internal group discussion and consultation during the advising meeting, we decided to remove Domain 1 (Tutors & After-school Programs) from our submission. It became clear that the idea did not align well with the expected depth and scale of the project and would not provide sufficient complexity for the assignment. As a result, we chose to discontinue this direction and introduce a new concept, which is presented later in the document under Domain 3.

~~1. **Domain Name #1 :** Tutors & After-school Programs~~

- ~~**The problem / Pain point:**~~

~~After-school tutoring centers face a complex scheduling challenge: each student requests different subjects, teachers have fixed availability, and rooms have limited capacity. Many students also enroll in multiple lessons, requiring careful coordination to avoid time conflicts.~~

~~These schedules are currently created manually using spreadsheets and manual coordination. As the number of students and subjects grows, this process becomes inefficient, error-prone, and difficult to maintain. Double bookings, overlapping lessons, and suboptimal room assignments are common, resulting in unnecessary administrative workload and reduced operational effectiveness.~~

- ~~**Target audience:**~~

~~The primary users affected by this problem are administrators and program coordinators working in private after-school tutoring centers. These are typically small to medium learning institutes that offer subject-based group and individual lessons outside regular school hours.~~

~~They are responsible for organizing student groups, assigning teachers, and managing room availability, and therefore experience the scheduling complexity and operational inefficiencies described above.~~

- ~~**Evidence / Discovery:**~~

~~To understand how after-school tutoring centers currently manage their operations, we spoke directly with an administrator and a program manager from a private learning center. They described a workflow that relies heavily on spreadsheets, manual coordination, and constant back-and-forth messaging. Both emphasized that creating the weekly schedule is the most demanding part of their job, particularly when trying to prevent students from being placed in overlapping lessons, ensuring that teachers are not double-booked, and managing room capacity. To complement~~

~~these conversations, we used ChatGPT to systematically explore online discussions, reviews, and practitioner insights about tutoring-center management. This broader scan revealed that many centers face the same issues: scheduling conflicts, inefficient manual processes, and frequent errors as student numbers grow. The combination of firsthand interviews and online evidence confirmed that scheduling complexity is a widespread and well-recognized pain poin~~t in this domain.

#	~~Evidence~~	~~Source, links~~
~~1~~	~~So far I've just been putting all of my scheduled sessions into a million different calendars to make sure I don't forget, and sending reminders to the parents the day before a lesson (I seem to get stood up a lot if I don't remind them lol)~~	~~link~~
~~2~~	~~I'm a boomer ELA tutor who never used Excel or Google sheets much/ever, but i'm kind of thinking I need something like that to keep track of my students' meeting dates, fees collected, what we did, and what I assigned~~	~~link~~
~~3~~	~~We currently log all our information through Excel under one master file. This info includes data such as past and upcoming lessons, student attendance, the amount paid to each tutor for each lesson, and client payment history.~~	~~link~~
~~4~~	~~Tutoring Center Scheduling Concerns~~	~~link~~
~~5~~	~~Managing a tutoring center without specialized software can be an uphill battle.~~ ~~Think about the everyday challenges: scheduling back-to-back tutoring sessions, tracking which students showed up (and which didn't), monitoring tutor availability, and compiling reports to show the impact of your services.~~	~~link~~

After reviewing the project requirements in more depth and discussing them together as a group, we decided to refine and adjust the concept for Domain 2. Our original framing did not fully align with the level of autonomy and multi-step decision-making expected from the AI agent. Therefore, we revised the domain to better fit the required scope and to ensure the agent can operate independently, perform several coordinated tasks, and provide meaningful value within the consulting context.
To achieve this, we expanded the agent's capabilities beyond simple financial organization and introduced three autonomous components that work together: an asset-discovery module that continuously scans stocks, crypto, ETFs, and startups based on user-defined preferences; a news-impact and price-prediction module that monitors financial news, classifies relevant events, and anticipates potential market movements; and a lightweight analytics module that applies basic quantitative checks, such as volatility estimates, trend strength, and similarity comparisons, to justify and validate the agent's recommendations. These new additions significantly broaden the system's functionality, making it a more complete and autonomous financial intelligence agent.

2. **Domain Name #2 :** Consulting

FINERIS - Financial Reasoning & Intelligence System

- **The problem / Pain point:**

In the consulting domain, many individuals want guidance not only on managing their spending and creating a clear saving plan, but also on understanding the wider financial environment that influences their decisions. While they often work with unorganized, text-based expense records such as notes, chat messages, or simple lists, they also face the challenge of navigating thousands of financial assets, ongoing market movements, and constant streams of news that may affect prices. Without a structured way to interpret personal spending patterns, discover suitable assets, evaluate the impact of financial news, or perform basic analytical checks, it becomes difficult for users to translate raw information into a practical and confident financial plan. The challenge is not the absence of data, users have expenses, market data is abundant, and news is continuous, but rather the effort required to connect these pieces and turn them into clear, personalized recommendations. This highlights the need for an autonomous, text-based assistant capable of interpreting unstructured

financial inputs, discovering relevant assets based on user-defined preferences, analyzing the impact of market events, and performing lightweight analytics to support well-reasoned financial decisions.

- **Target audience:**

The people who face this problem are everyday individuals who try to manage their personal finances using simple notes, messages, or lists, and want help turning this information into a clear saving plan. At the same time, they often struggle to make sense of the broader financial environment, such as choosing suitable assets, interpreting market news, or understanding basic indicators, that could support their financial goals. These users already have access to the data, but they need an autonomous system that can organize it, connect it with relevant market insights, and provide clear, practical guidance to improve their financial decisions.

- **Evidence / Discovery:**

When considering possible AI agents, this idea surfaced immediately because managing personal spending and creating a saving plan is a challenge many people face. To validate that this is a widespread issue, we used ChatGPT to search through Reddit discussions and online conversations where people openly talk about their difficulties with budgeting and saving. These posts show consistent patterns: people are unsure how to organize their spending data, how to identify areas for improvement, and how to build a realistic saving plan. The evidence confirms that this is a real and common pain point that impacts a broad population.

#	Evidence	Source, links
1	Struggling to save money and have no clue where to begin	link
2	I want to get serious about paying off my debt and building savings, but I don't know where to start or how to keep myself from overspending.	link
3	Any tips for someone who struggles to save?	link

- **Data sources:**

1-

- Data source name: Stock Market Dataset
- Data source type: RAG / CSV
- Data source link: https://www.kaggle.com/datasets/jacksoncrow/stock-market-dataset
- Data source size: 548 MB (ZIP download)
- Data source description:

  This dataset contains daily historical stock price data for hundreds of publicly traded companies over multiple years. It includes fields such as open, high, low, close, adjusted close, and trade volume. This information is essential for computing volatility, trend strength, similarity patterns, and other lightweight mathematical indicators. These analytical features directly support the autonomous decision-making of the Lightweight Analytics Domain, enabling the agent to justify recommendations and evaluate asset behavior based on real market dynamics.
- Data source table example:

Data	Open	High	Low	Close	Adj Close	Volume	Name
2013-02-08	15.07	15.12	14.63	14.78	11.76	24,249,500	AAPL
2013-02-11	14.89	14.93	14.56	14.67	11.68	18,256,100	AAPL
2013-02-12	14.62	14.76	14.56	14.65	11.66	15,235,700	AAPL

2-

- Data source name: Yahoo Finance Search API (News & Asset Metadata)
- Data source type: RAG / REST API
- Data source link:

  - General search: https://query1.finance.yahoo.com/v1/finance/search
  - Example query (company-specific news): https://query1.finance.yahoo.com/v1/finance/search?q=nvidia

- Data source size: Not a fixed dataset. dynamic API returning lightweight JSON responses (typically 5–50 KB per request)
- Data source description:

  This API provides real-time financial information, including news articles, sentiment-related metadata, asset descriptions, ticker matches, sectors, and relevance scores. It is essential for the News Impact & Price Prediction Domain,

enabling the agent to continuously retrieve and analyze finance-related news events. The agent uses this data to classify sentiment, detect important triggers (e.g., earnings, partnerships, lawsuits), and anticipate potential price reactions. The API also supports the Asset Discovery Domain by providing metadata about companies, sectors, and related assets, helping the system filter opportunities based on user-defined criteria.

3-

- Data source name: Synthetic User Financial Profiles Dataset
- Data source type: Synthetic / Postgres / Internal
- Data source description:

  This synthetic dataset will be generated to represent the financial profiles and preferences of different users interacting with the autonomous agent. It includes fields such as risk tolerance, investment horizon, starting budget, asset preferences, prohibited sectors, and saving goals. These attributes enable the agent to personalize its recommendations and align decisions from all three domains (asset discovery, news-impact reasoning, and analytics) with user constraints. Since the assignment requires autonomous behavior, the synthetic profiles allow us to simulate realistic scenarios, evaluate the agent across diverse risk categories, and test its ability to make individualized decisions without relying on private or sensitive real-world user data.
- Data source table example:

UserID	RiskLevel	BudgetUSD	TimeHorizon (months)	PrefferedSectors	ExcludedSectors	StartingAmount	SavingGoal
U001	Medium	800	6	Tech, Healthcare	Energy	500	2000
U002	High	1500	12	Crypto, AI	Banking	1000	5000
U003	Low	400	3	ETFs, Consumer Goods	Crypto	300	1000

3. **Domain Name #3 :** Commercial Real Estate

PROPCORE - Property Commercial Operations & Reasoning Engine

- **The problem / Pain point:**

In the commercial real estate sector, matching businesses with suitable properties is a complex and time-consuming process. Each company has highly specific operational needs, such as required floor space, zoning restrictions, power capacity, foot-traffic demands, lease duration, and budget constraints, that must be aligned with property characteristics that vary widely across locations. Agents often manage hundreds of listings and inquiries simultaneously, while property owners expect timely tenant placement and minimal vacant periods. Because much of the information arrives as unstructured text (emails, descriptions, business requirements, property notes), it becomes difficult to evaluate compatibility, identify strong matches, and maintain momentum across many ongoing negotiations. Without an autonomous system that can parse requirements, analyze property attributes, detect stalled listings, and proactively recommend next actions, the commercial leasing workflow remains inefficient, reactive, and prone to missed opportunities.

- **Target audience:**

The primary users who experience this problem are commercial real estate brokers, property managers, and leasing coordinators who handle large portfolios of offices, retail units, industrial spaces, and mixed-use properties. These professionals are responsible for evaluating incoming tenant requirements, maintaining active listings, and ensuring that vacancies are filled efficiently. They rely heavily on text-based communication with clients and property owners, often receiving unstructured business requirements that must be matched to a diverse and constantly changing inventory of commercial spaces. This audience benefits directly from an autonomous system that can interpret requirements, identify suitable properties, and support the leasing workflow more proactively.

- **Evidence / Discovery:**

When exploring potential directions for an autonomous agent within the commercial real estate domain, it quickly became clear that brokers and leasing teams face recurring challenges that could benefit from automation. To validate this observation, we used ChatGPT to scan discussions, forums, and industry conversations where agents describe the day-to-day difficulties of matching businesses with appropriate

commercial spaces. These posts consistently highlighted issues such as managing large volumes of unstructured tenant requirements, struggling to compare diverse property attributes, and dealing with listings that remain inactive due to slow turnaround or missed follow-ups. The evidence shows a clear pattern: commercial real estate professionals operate in an environment where information is fragmented, coordination is manual, and opportunities can easily be lost without timely, data-driven support. This confirms that the pain points addressed by our proposed agent are real and widely experienced in this industry.

#	Evidence	Source, links
1	"it is generally hard to find a tenant, there are already a ton of commercial spaces up for lease; competition is high and finding reliable tenants is difficult." (comment: "If you have to ask on reddit, maybe you aren't ready for this. … In general you will have a hard time finding a tenant.")	link
2	"Most commercial landlords are struggling to find tenants for their retail or office spaces." (post begins: "I have a few spaces available … but am having trouble finding anyone that actually wants to rent.")	link
3	"Today's biggest problem in commercial real estate is the rising number of empty buildings, many commercial properties remain vacant, causing significant losses for owners." This article expresses a widespread challenge of high vacancy rates.	link

- **Data sources:**

1-

- Data source name: Commercial Real Estate for Sale
- Data source type: RAG / CSV
- Data source link: https://www.kaggle.com/datasets/aramacus/commercial-real-estate-for-sale
- Data source size: 1.77 MB
- Data source description:

  This dataset contains real commercial property listings, including offices, retail spaces, industrial units, warehouses, and mixed-use buildings. Each entry includes a combination of structured fields, such as location, price, property type, size, numerical attributes, along with unstructured text descriptions written by property sellers. This pairing of structured and free-text data makes the dataset well-suited for our autonomous AI agent, which

must parse property descriptions, extract relevant features, match tenant requirements to property attributes, and identify suitable spaces for different business needs. The dataset enables tasks like semantic matching, constraint-based filtering, vacancy detection, and ranking of properties based on business requirements

- Data source table example:

Title	Price	Nbn	Address	Text	Area	Type	Latitude	Longitude
Premium Office Space – CBD Location	1200000	Yes	155 George St, Sydney NSW	Modern office suite with multiple rooms, conference space, and direct access to public transport	320	Office	-33.8650	151.2094
Retail Space in High Foot-Traffic Area	850000	Yes	78 Collins St, Melbourne VIC	Ideal for cafés, boutique stores, or service-based businesses	150	Retail	-37.8136	144.9631
Large Industrial Warehouse	2400000	No	12 Barwon Terrace, Geelong VIC	Warehouse with loading docks, wide access, and large open floor space suitable for logistics	1100	Industrial	-38.1500	144.3500

2-

- Data source name: Synthetic Dataset for Tenants, Calendars, and Communication Logs
- Data source type: Synthetic / Postgres / Internal
- Data source description:

  To effectively evaluate the autonomous behavior of the commercial real estate agent, we generate synthetic data representing potential tenants, scheduling calendars, and communication logs. This dataset allows us to simulate realistic interactions between businesses searching for commercial properties and the agent's matching, scheduling, and coordination processes. The synthetic profiles include business requirements, preferred property features, budget ranges, industry types, and operational constraints. Calendars simulate property viewing availability and agent time slots, while communication logs resemble the email- or message-based requests that agents typically receive in unstructured form. This synthetic dataset enables controlled testing of matching logic, scheduling decisions, and multi-agent coordination across a wide range of scenarios without relying on private or sensitive real-world data.